

Infix to Postfix Conversion

Rules :-

- When an operand is read, it is immediately placed onto the output
- When an operator is seen, it is placed on the stack, Stack represents pending operators
- High precedence operator if present should be popped out before placing the current operator.

① $a + b * c + (d * e + f) * g$

Infix character scanned	Stack	Postfix expression
a	-	a
+	+	a
b	+	ab
*	+ *	ab
c	+ *	abc
+	+	abc*+
(	+(	abc*+
d	+(	abc*+d
*	+(*	abc*+d
e	+(*	abc*+de
+	+(+	abc*+de*
f	+(+	abc*+de*f
)	+	abc*+de*f+
*	+ *	abc*+de*f+
g	+ *	abc*+de*f+g

Ans $\rightarrow abc*+de*f+g*+$

$$(2) (A - (B / C + (D \% E * F) / G) * H)$$

Infix character	Stack	Postfix
(	(	-
A	(	A
-	(-	A
(	(- (	A
B	(- (	AB
/	(- (/	AB
C	(- (/	ABC
+	(- (+	ABC/
(	(- (+ (	ABC/
D	(- (+ (	ABC/D
%	(- (+ (%	ABC/D
E	(- (+ (%	ABC/DE
*	(- (+ (*	ABC/DE%
F	(- (+ (*	ABC/DE%*F
)	(- (+	ABC/DE%*F*
/	(- (+/	ABC/DE%*F*

G	(- (+ /	ABC / DE * % F * G
)	(-	ABC / DE * % F * G / +
*	(- *	ABC / DE * % F * G / +
H	(- *	ABC / DE * % F * G / + H
)	(- *)	ABC / DE * % F * G / + H * -

Expression evaluation

→ Postfix evaluation

→ Prefix evaluation

POSTFIX EVALUATION

Rules :-

- (i) When a number is seen, it is pushed onto the stack
- (ii) When an operator is seen, the operator is applied to the two numbers (symbols) that are popped from the stack
- (iii) Do the operation and the result is pushed into the stack.

Example:-

Evaluate the postfix expression

6 5 2 3 + 8 * + 3 + *

Sol:-

Symbol	Stack
6	Push "6" 6
5	Push "5" 5 6
2	Push "2" 2 5 6
3	Push "3" 3 2 5 6
+	Pop two operands Push (A) = 3 Push (B) = 2 The operation must be performed like (B+A). B → A order 2+3=5 ↓ 5 5 6

8

Push "8"

8
5
5
6

*

$$5 * 8 = 40$$

40
5
6

+

$$5 + 40 = 45$$

45
6

3

Push "3"

3
45
6

+

$$45 + 3 = 48$$

48
6

*

$$6 * 48 = 288$$

288

Example 2: Evaluate the postfix expression

7 5 3 * 5 1 ^ / + 3 2 - +

Sol:

Symbol	Stack
7	7
5	5 7
3	3 5 7
*	$5 * 3 = 15$ 15 7
5	5 15 7
1	1 5 15 7
^	$\begin{matrix} \nearrow 5 \wedge 1 \\ B \quad A \end{matrix} = 5$ 5 15 7

1

$$\frac{15}{5} = 3$$

$$\begin{array}{r} 3 \\ \hline 7 \end{array}$$

+

$$7 + 3 = 10$$

$$\begin{array}{r} 10 \\ \hline \end{array}$$

3

$$\begin{array}{r} 3 \\ \hline 10 \end{array}$$

2

$$\begin{array}{r} 2 \\ \hline 3 \\ \hline 10 \end{array}$$

-

$$3 - 2 = 1$$

$$\begin{array}{r} 1 \\ \hline 10 \end{array}$$

+

$$10 + 1 = 11$$

$$\begin{array}{r} 11 \\ \hline \end{array}$$

Example 3 :- Evaluate the postfix expression

5 3 + 8 2 - *

Sol:

Symbol	Stack
5	5
3	3 5
+	8
8	8 8
2	2 8 8
-	8 - 2 = 6 6 8
*	8 * 6 = 48 48

PREFIX EVALUATION

→ Reverse the expression

→ $\begin{array}{|c|} \hline A \\ \hline B \\ \hline \end{array} \downarrow$ ^{e.g.} $A+B$. Evaluated by taking top operand A and then B
Bottom operand B

Example 1 :- Evaluate the prefix notation

$+ , - , * , 2 , 2 , / , 16 , 8 , 5$

Solution :

Reverse the expression $\rightarrow 5, 8, 16, /, 2, 2, *, -, +$

Symbol		Stack
5		$\begin{array}{ c } \hline 5 \\ \hline \end{array}$
8		$\begin{array}{ c } \hline 8 \\ \hline 5 \\ \hline \end{array}$
16		$\begin{array}{ c } \hline 16 \\ \hline 8 \\ \hline 5 \\ \hline \end{array}$
/	$16/8=2$	$\begin{array}{ c } \hline 2 \\ \hline 5 \\ \hline \end{array}$
2, 2		$\begin{array}{ c } \hline 2 \\ \hline 2 \\ \hline 5 \\ \hline \end{array}$
*	$2*2=4$	$\begin{array}{ c } \hline 4 \\ \hline 5 \\ \hline \end{array}$
-	$4-2=2$	$\begin{array}{ c } \hline 2 \\ \hline 5 \\ \hline \end{array}$
+	$2+5$	$\begin{array}{ c } \hline 7 \\ \hline \end{array}$

Example 2: Evaluate the prefix expression.

$-, +, 2, *, 3, 4, /, 16, \wedge, 2, 3$

Sol:

Reverse $\rightarrow 3, 2, \wedge, 16, /, 4, 3, *, 2, +, -$

Symbol	Stack
3	$\boxed{3}$
2	$\boxed{\begin{array}{c} 2 \\ 3 \end{array}}$
$\wedge$	Here, $A \rightarrow 2,$ $B \rightarrow 3$ It follows $A \rightarrow B$ order. $2^3 = 8$ $\boxed{8}$
16	$\boxed{\begin{array}{c} 16 \\ 8 \end{array}}$
/	$A/B = 16/8 = 2$ $\boxed{2}$
4	$\boxed{\begin{array}{c} 4 \\ 2 \end{array}}$
3	$\boxed{\begin{array}{c} 3 \\ 4 \\ 2 \end{array}}$
*	$3 * 4 = 12$ $\boxed{\begin{array}{c} 12 \\ 2 \end{array}}$

2

2
12
2

+

$$2 + 12 = 14$$

14
2

-

$$14 - 2 = 12$$

12

Example 3 : Evaluate the prefix expression.

$$+ 9 * 2 6$$

Sol :

Reverse $\rightarrow 6 2 * 9 +$

Symbol

Stack

6

6

2

2
6

*

$$2 * 6 = 12$$

12

9

9
12

+

$$9 + 12 = 21$$

21

Infix to Prefix conversion

Rules:

- (i) Reverse the infix expression
- (ii) Obtain the "nearly" postfix expression of the modified expression
- (iii) Reverse the postfix expression.

Example 1 :- $(A + B \wedge C) * D + E \wedge 5$

Sol:-

Reverse $\rightarrow 5 \wedge E + D *) C \wedge B + A ($

Input	Stack	Output
5		5
^	^	5
E	^	SE
+	+	SE^
D	+	SEAD
*	+*	SEAD
)	+*)	SEAD
C	+*)	SEADC
^	+*)^	SEADC
B	+*)^	SEADCB

+	+	*)	+	SEADCB
A	+	*)	+	SEADCBAA
1	+	*)		SEADCBAA+

SEADCBAA+*+

Reverse $\rightarrow$ +*+AABCDIES

Rules:

- (i) First, reverse the infix expression given in the problem
- (ii) Scan the expression from left to right
- (iii) Whenever operands arrive, print them.
- (iv) If the operation arrives & the stack is found to be empty, then simply push the operators into the stack.
- (v) If the incoming operator has higher precedence than the top of the stack, push the incoming operator into the stack
- (vi) If the incoming operator has the same precedence with a top of the stack, push the incoming operator into the stack.
- (vii) If the incoming operator has lower precedence than the top of the stack, pop, and print the top of the stack. Test the incoming operator against the top of the stack again & pop the operator from the stack till it finds the operator of a lower precedence or same precedence.
- (viii) If the incoming operator has the same precedence with the top of the stack & the incoming operator is 1, then pop the top of the stack till the condition is true. If the condition is not true, push the 1 operator

Rules (continued) :-

- (ix) When we reach the end of the expression, pop, and print all the operators from the top of the stack.
- (x) If the operator is ')', then push it into the stack.
- (xi) If the operator is '(', then pop all the operators from the stack till it finds ')' opening bracket in the stack.
- (xii) If the top of the stack is ')', push the operator on the stack.
- (xiii) At the end, reverse the output.

Example 2: $K + L - M * N + (O \wedge P) * W / U / V * T + Q$

Sol: Reverse $\rightarrow Q + T * V / U / W *) P \wedge O (+ N * M - L + K$

Symbol	Stack	Prefix
Q		Q
+	+	Q
T	+	QT
*	+	QT
V	+	QTV
/	+	QTV
U	+	QTVU
/	+	QTVU
W	+	QTVUW
*	+	QTVUW
)	+	QTVUW
P	+	QTVUWP
$\wedge$	+	QTVUWP
O	+	QTVUWPO
(	+	QTVUWPO
+	+	QTVUWPO
N	+	QTVUWPO

*

++*

QTVUWPOΛ*//*N

M

++*

QTVUWPOΛ*//*NM

-

++-

QTVUWPOΛ*//*NM*

L

++-

QTVUWPOΛ*//*NM*L

+

++-+.

QTVUWPOΛ*//*NM*L

K

++-+

QTVUWPOΛ*//*NM*LK

QTVUWPOΛ*//*NM*LK+-++

Reverse → ++-+KL*MN*//*ΛOPWUVTQ

Eg 3: infix to prefix

Convert infix to prefix

$$A + B * C / D$$

Sol: Reverse $\rightarrow D / C * B + A$

Symbol	Stack	prefix
D		D
/	/	D /
C	/ C	D C /
*	/ * C	D C * /
B	/ * B C	D C B * /
+	+ / * B C	D C B * / +
A	+ A / * B C	D C B * / + A

D C B * / + A

Reverse $\rightarrow + A / * B C D$

Eg 4: $A \wedge B * C / (D * E - F)$

Sol:

Reverse $\rightarrow$) F - E * D (/ C * B $\wedge$ A

Symbol	Stack	Prefix
)	)	
F	)	F
-	) -	F -
E	) -	F E
*	) - *	F E *
D	) - * D	F E D
(		F E D * -
/	/	F E D * - /
C	/	F E D * - C
*	/ *	F E D * - C *
B	/ *	F E D * - C B
$\wedge$	/ * $\wedge$	F E D * - C B $\wedge$
A	/ * $\wedge$ A	F E D * - C B A
F E D * - C B $\wedge$ A * /		

Reverse $\rightarrow$ / * $\wedge$ A B C - * D E F

Ex: 5: Convert infix to prefix

$$(A+B) * (C-D) + X \wedge Y \wedge Z * T$$

Solution: Reverse $T * Z \wedge Y \wedge X +) D - C (*) B + A ($

Symbol	Stack	Prefix
T		T
*	*	T
Z	*	TZ
$\wedge$	* $\wedge$	TZ
Y	* $\wedge$	TZY
$\wedge$	* $\wedge$	TZY $\wedge$
X	* $\wedge$	TZY $\wedge$ X
+	+	TZY $\wedge$ X $\wedge$ +
)	+))	TZY $\wedge$ X $\wedge$ +
D	+))	TZY $\wedge$ X $\wedge$ +
-	+) -	TZY $\wedge$ X $\wedge$ +
C	+) -	TZY $\wedge$ X $\wedge$ +
(	+) - ($\rightarrow$ POP	TZY $\wedge$ X $\wedge$ +
*	+*	TZY $\wedge$ X $\wedge$ +
)	+*))	TZY $\wedge$ X $\wedge$ +
B	+*))	TZY $\wedge$ X $\wedge$ +
+	+*) +	TZY $\wedge$ X $\wedge$ +
A	+*) +	TZY $\wedge$ X $\wedge$ +

$(\mid + *) + (\rightarrow \text{POP} \mid$

TZYAXAX*DC-BA+

TZYAXAX*DC-BA+ * +

Reverse $\rightarrow + * + AB - CD * AX \wedge YZT$